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Distributionally Robust Coordinated Scheduling of Power-Transportation Systems Considering Bounded Rationality and Flexible Swapping of Electric Vehicles

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ABSTRACT

Large-scale electric vehicles (EVs) integration demand coordinated power-transportation systems to optimise charging loads and grid flexibility, whereas existing strategies inadequately address range anxiety, peak demands and photovoltaic (PV) operational uncertainties. To address these issues, this work proposes a power-transportation coordination model integrating flexible swapping. The model strategically guides EV users towards swapping stations by embedding bounded rationality parameters within a stochastic user equilibrium framework, mitigating anxiety while balancing charging-swapping loads. This work also develops a distributionally robust optimisation framework to address PV uncertainty, enhancing resilience against generation fluctuations. Numerical simulations demonstrate that battery swapping integration significantly reduces distribution network peak loads and shortens user waiting times compared to charging-only approaches. The proposed method outperforms deterministic models by reducing conservativeness and aligning more closely with real-world operational dynamics, validating its efficacy in harmonising user behaviour and grid constraints under uncertainty.

1 | Introduction

The increasing penetration of EVs has intensified operational interdependencies between PDN and TN, necessitating coordinated power-transportation management [1, 2]. Although BSS offers a strategic solution to mitigate range anxiety and diversify energy replenishment modes [3, 4], surging EV adoption imposes dual pressures: spatially heterogeneous charging demands and temporally concentrated peak loads. Disorderly charging patterns exacerbate peak load stresses on PDN, risking voltage violations and equipment overloads [5, 6], whereas urban–rural charging load imbalances trigger severe congestion at urban stations during peak periods degrading user satisfaction [7].

Our work proposes a coordination power-transportation model integrating BSS, enabling adaptive load management through spatiotemporal charge-swapping optimisation. This model not only balances PDN loads by redistributing energy demands but also enhances user experience via obvious queueing time reduction, ultimately advancing the economic and operational efficiency of integrated power-transportation systems. Besides, BSSs also serve as flexible loads and energy storage units for the grid. They allow for the decoupling of power demand from vehicle arrival times, thereby providing critical ancillary services such as peak shaving and valley filling.

The technical literature includes works on the design and operation of the coordination model. Early studies, such as those by Li et al. [8] and Baghali et al. [9], established foundational models under static traffic flow assumptions, demonstrating the efficacy of integrated systems in energy conservation and load management. Subsequent work by Lv et al. [10] introduced temporal dynamics through multiperiod optimisation, addressing time-varying demands in semidynamic

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frameworks. Further advancements by Zhou et al. [11] and Xie et al. [12] incorporated dynamic traffic assignment models to capture spatiotemporal interactions between power and traffic flows, whereas this work focuses on the static models to increase calculating speed and simplify coordination model. Operational enhancements were achieved through infrastructure planning, economic incentives and demand forecasting. Key implementations include Zhang et al. [13] optimising EV routing via second-order cone programming, K. Li et al. [14] designing profit-driven pricing models, Cui et al. [15] applying Frank-Wolfe algorithms for traffic flow optimisation and Wang et al. [16] proposing a planning model to seek to optimise distribution system economics by minimising operational expenditures while maximising annual traffic throughput.

Other related works focus on the critical challenges such as uncertainty management, privacy preservation and dynamic pricing. P. Li et al. [17] proposed a robust reinforcement learning method to mitigate uncertainties in EV charging behaviour while safeguarding data privacy. Shen et al. [18] further enhanced congestion management in PDN through distributed dynamic tariffs tailored to EV spatiotemporal coordination, whereas this work introduces the SUE model to achieve the QRE state, successfully coordinate the congestion management in PDN. For renewable-integrated systems, Lin et al. [19] developed robust optimisation frameworks to optimise PV-storage operations and EV charging load management, whereas this work introduces DRO algorithm to reduce the degree of conservation relatively and improve the calculating accuracy. Moreover, artificial intelligence methods have achieved remarkable results in coordination systems. Ding et al. [20] proposes a probabilistic forecasting framework using Bayesian Neural Networks and Transformer models to predict the flexible regulation capability of electric vehicles and Qiu et al. [21] introduced a graph-based multiagent reinforcement learning approach to optimise the joint routing and charging scheduling of EVs in coupled power-transportation networks.

Although these studies have significantly advanced the mitigation of users' range anxiety and the coordination of power-transportation operations, key limitations persist. First, most research focuses solely on charging stations as the primary units for optimisation, overlooking the critical role of battery swapping stations. This neglect exacerbates grid security risks posed by escalating charging demands. Second, prevailing models remain largely deterministic, ignoring uncertainties in distributed generation output, which compromises their robustness and real-world applicability. Furthermore, simplified user decision-making frameworks fail to account for bounded rationality in charging/swapping behaviours and computational scalability remains constrained by nonconvex user equilibrium formulations. These limitations underscore the need for innovative coordination mechanisms that integrate battery swapping dynamics and adaptive optimisation under uncertainty.

To bridge these limitations above, we propose a power-transportation coordination framework that integrates flexible swapping and bounded rationality, which employ the SUE model to determine the probability distribution of user choices for charging or swapping under varying BRPs. Unlike deterministic choice or deterministic user equilibrium, which implicitly assumes perfectly rational users with complete information and often leads to an extreme all-or-nothing switch when costs are close, the logit-based SUE captures EV users' bounded rationality and perception errors regarding waiting time, detour distance and charging/swapping prices. This probabilistic choice formulation produces more realistic demand splitting across charging and swapping options and has been widely used in transportation behaviour and equilibrium modelling [22]. Moreover, we introduce DRO algorithm to optimise the coordination model under renewable energy uncertainty, further increasing the robustness of daily operation.

Using two test systems that represent current power-transportation systems, we demonstrate the importance of introducing the flexible swapping and user bounded rationality. We also investigate the interdependency of BRPs on charging/swapping behaviour by adjusting BRPs from zero to one.

The remainder of this paper is organised as follows. Sections 2 and 3 elaborate the mathematical formulation. Sections 4 and 5 present numerical results of two test systems. Section 6 concludes.

2 | Model Formulation

Figure 1 presents the proposed coordination model for PDN and TN. This framework comprises two distinct layers: the top layer represents the TN, and the bottom layer corresponds to the PDN and the charging-swapping stations function as the coupling unit. As can be seen, the PDN calculates the power flow and modifies the charging price according to the real-time charging load, whereas the TN adjusts the distribution of charging EVs according to the real-time charging price.

The following common assumptions are made in our work:

1. The TN model assumes a static topology, excluding real-time reconfiguration during operation. Although real-world traffic is dynamic, this study focuses on hourly level energy scheduling. The static topology assumption provides a sufficient macroscopic approximation of traffic flows for energy exchange analysis, properly balancing modelling accuracy and computational complexity.
2. The coupling unit integrates multiple CSSs, serving as the interface between the transportation and distribution networks.
3. In the small-scale TN under study, all EVs are assumed to possess battery swapping capability.

2.1 | EV Charging or Battery Swapping Decision-Making Model

Figure 2 illustrates the structure of the CSS, which consists of multiple BSSs and FCSs. The BSS offers battery swapping services for EVs, whereas the FCS provides rapid charging services. Both BSS and FCS employ efficient battery charging strategies, ensuring that depleted batteries are promptly sent to charging stations for recharging, whereas fully charged batteries are

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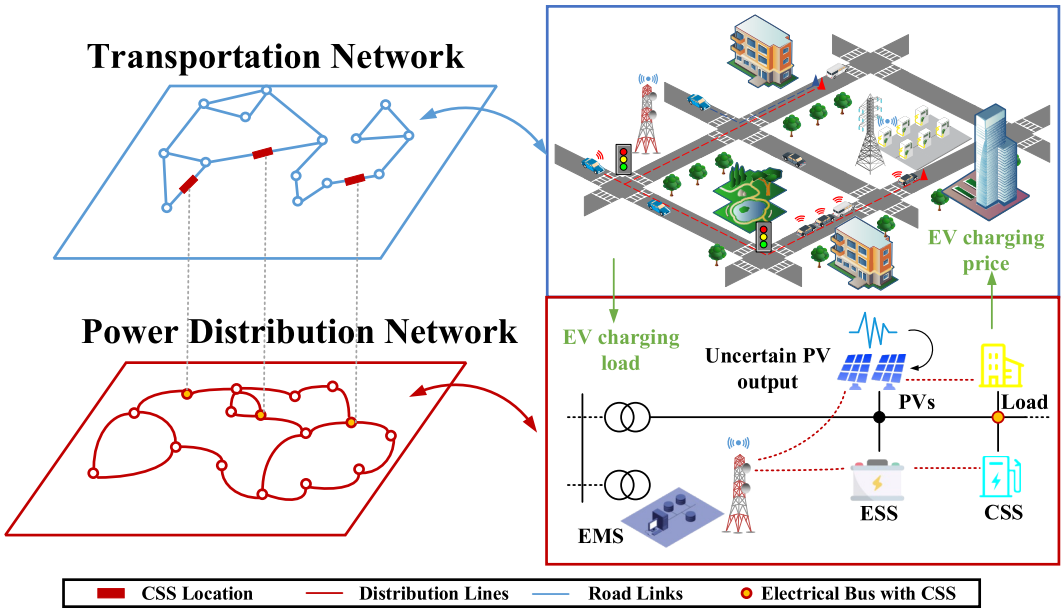


FIGURE 1 | Structure of the proposed model.

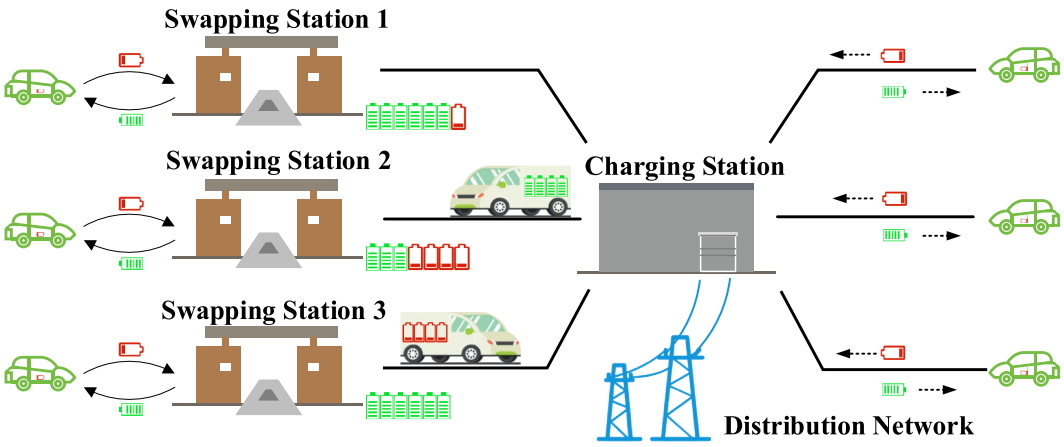


FIGURE 2 | Cooperative operation framework of the CSS.

transported to BSS to maintain the seamless operation of the integrated stations [23].

To enhance the accuracy of charging and swapping decisions, EV users are categorised into two groups: those requiring charging, denoted as N_{ch} and those opting for battery swapping, denoted as N_{sp}. The total number of EVs denoted as N_{EV}.They are calculated as follows:

$$N_{\mathrm{EV}} = N_{\mathrm{ch}} + N_{\mathrm{sp}} \tag{1}$$

$$N_{\mathrm{ch}} = \pi_{\mathrm{ch}} N_{\mathrm{EV}} \tag{2}$$

$$N_{\mathrm{sp}} = \pi_{\mathrm{sp}} N_{\mathrm{EV}} \tag{3}$$

$$\pi_{\mathrm{EV}} = \pi_{\mathrm{ch}} + \pi_{\mathrm{sp}} \tag{4}$$

The total charging time for a single EV in the FCS can be expressed using the M/M/c/K model [24], as shown in Equation (5), whereas the total charging cost for the EV is shown in Equation (6). Similarly, the total swapping time of an EV can be expressed as Equation (7) and the total swapping cost can be expressed as Equation (8).

$$t_a^{\mathrm{ch}} = \frac{E}{p} + t_{a,\mathrm{ch}}^{\max}\left(\frac{N_{a,\mathrm{ch}}}{c_a^{\mathrm{ch}}}\right)^3 : N_{a,\mathrm{ch}} \le c_a^{\mathrm{ch}}, a \in T_{\mathrm{A}}^{\mathrm{ch}} \tag{5}$$

$$C^{\mathrm{ch}} = \omega\left(t_a^{\mathrm{rg}} + t_a^{\mathrm{ch}}\right) + c_{e,t}E + M^{\mathrm{ch}} \tag{6}$$

$$t_a^{\mathrm{sp}} = t^{\mathrm{sp}} + t_{a,\mathrm{sp}}^{\max}\left(\frac{N_{a,\mathrm{sp}}}{c_a^{\mathrm{sp}}}\right)^3 : N_{a,\mathrm{sp}} \le c_a^{\mathrm{sp}}, a \in T_{\mathrm{A}}^{\mathrm{sp}} \tag{7}$$

$$C^{\mathrm{sp}} = \omega\left(t_a^{\mathrm{rg}} + t_a^{\mathrm{sp}}\right) + c_{e,t}E + M^{\mathrm{sp}} \tag{8}$$

Equations (1–4) are used to calculate the number of EVs in different groups with Equation (1) representing the total number of EVs, and Equations (2) and (3) indicating the number of EVs opting for charging and battery swapping, respectively. Equation (4) calculates the EV penetration rate. Building on this, Equations (5–8) model the operational aspects of EVs where Equations (5) and (7) represent the total charging or swapping time at the FCS and BSS, respectively, whereas

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Equations (6) and (8) calculate the corresponding charging and battery swapping costs for a single EV.

In the actual charging process, EV users are influenced by both charging and swapping costs as well as time costs. When users are pressed for time, they tend to choose the battery swapping mode, which typically incurs higher energy replenishment costs. Conversely, when users have more time, they are more likely to choose the charging mode, which generally has a lower cost.

This study employs the SUE model to analyse EV users' charging and swapping behaviours. Under the SUE framework, EVs self-dispatch based on rational decision-making to minimise their perceived generalised cost. The proposed coordination mechanism mitigates network congestion and optimises charging prices, thereby reducing the aggregate service time and cost, which inherently preserves and improves the social welfare of EV users.

The probability of selecting a charging station is denoted as $\alpha_{\mathrm{ch},n}$, whereas the probability of choosing a BSS is represented by $\alpha_{\mathrm{sp},n}$. To account for potential user errors in cost perception, random cognitive errors for charging and swapping are defined as $\varepsilon_{\mathrm{ch},n}$ and $\varepsilon_{\mathrm{sp},n}$, respectively. Based on this, the random charging cost $\tilde{C}_{\mathrm{ch}}(\alpha_{\mathrm{ch},n})$ and battery swapping cost $\tilde{C}_{\mathrm{sp}}(\alpha_{\mathrm{sp},n})$ can be expressed as follows:

$$\tilde{C}_{\mathrm{ch}}(\alpha_{\mathrm{ch},n}) = C_{\mathrm{ch}}(\alpha_{\mathrm{ch},n}) + \varepsilon_{\mathrm{ch},n} \tag{9}$$

$$\tilde{C}_{\mathrm{sp}}(\alpha_{\mathrm{sp},n}) = C_{\mathrm{sp}}(\alpha_{\mathrm{sp},n}) + \varepsilon_{\mathrm{sp},n} \tag{10}$$

Assume that the random terms $\varepsilon_{\mathrm{ch},n}$ and $\varepsilon_{\mathrm{sp},n}$ follow independent Gumbel distributions with their cumulative distribution function (CDF) given by the following form [25].

$$F(x) = \Pr[\varepsilon_{\mathrm{ch/sp},n} \le x] = \mathrm{e}^{-\mathrm{e}^{-\lambda x}} : \lambda \ge 0 \tag{11}$$

The derivation results in the polynomial Logit form of the QRE as shown in Equation (12). Under the QRE framework, the user's charging and swapping decisions are expressed as follows [26] where the parameter λ serves as the scale parameter reflecting the user's cognitive precision regarding costs referred to as the BRP. A larger λ corresponds to a smaller perception error variance and higher rationality, whereas $\lambda \to 0$ implies completely random decision-making.

Besides, we assume a homogeneous λ to capture the aggregate behaviour of the EV fleet. Although individual heterogeneity exists, a representative agent approach is adopted to maintain the tractability of the coupled optimisation model.

$$\alpha_{\mathrm{ch},n} = \frac{\mathrm{e}^{-\lambda C_{\mathrm{ch}}}}{\mathrm{e}^{-\lambda C_{\mathrm{ch}}} + \mathrm{e}^{-\lambda C_{\mathrm{sp}}}} \tag{12}$$

$$\alpha_{\mathrm{sp},n} = 1 - \alpha_{\mathrm{ch},n} \tag{13}$$

The equations above describe the decision-making process in the charging and swapping behaviour. Equations (9) and (10) represent the random charging cost and random battery swapping cost, respectively. Equation (11) shows the CDF of the Gumbel distribution, whereas Equations (12) and (13) provide the probability calculation formulae for choosing the FCS and the BSS, respectively.

2.2 | PDN Model Considering BRP

The model of PDN considering BRP is as follows [27]:

$$\min C_{\mathrm{PDN}} = C_{\mathrm{sub}} = \sum_{t=1}^{T} c_{\mathrm{sub},t} P_{\mathrm{sub},t} \tag{14}$$

$$\text{s.t.} P_{ij,t} + P_{j,t}^{\mathrm{PV}} = \sum_{k \in N(j)} P_{jk,t} + P_{j,t}^{\mathrm{load}} + r_{ij} l_{ij,t} \tag{15}$$

$$Q_{ij,t} + Q_{j,t}^{\mathrm{PV}} = \sum_{k \in N(j)} Q_{jk,t} + P_{j,t}^{\mathrm{load}} \tan\theta_j + x_{ij} l_{ij,t} \tag{16}$$

$$V_{j,t} = V_{i,t} - 2\left(P_{ij,t} r_{ij} + Q_{ij,t} x_{ij}\right) + l_{ij,t}\left(r_{ij}^2 + x_{ij}^2\right) \tag{17}$$

$$l_{ij,t} V_{ij,t} \ge P_{ij,t}^2 + Q_{ij,t}^2 \tag{18}$$

$$U_{i,\min} \le U_{i,t} \le U_{i,\max} \tag{19}$$

$$I_{ij,\min} \le I_{ij,t} \le I_{ij,\max} \tag{20}$$

$$P_{ij,t}^2 + Q_{ij,t}^2 \le S_{ij,\max}^2 \tag{21}$$

$$P_{\mathrm{sub}}^{\min} \le P_{\mathrm{sub},t} \le P_{\mathrm{sub}}^{\max} \tag{22}$$

$$Q_{\mathrm{sub}}^{\min} \le Q_{\mathrm{sub},t} \le Q_{\mathrm{sub}}^{\max} \tag{23}$$

$$Q_{\mathrm{PV}}^{\min} \le Q_{\mathrm{PV},j,t} \le Q_{\mathrm{PV}}^{\max} \tag{24}$$

$$P_{\mathrm{PV},j,t}^2 + Q_{\mathrm{PV},j,t}^2 \le \left(S_{\mathrm{PV}}^{\max}\right)^2 \tag{25}$$

Here, the AC power flow model is built in Equations (15–18), which include the power balance constraints in (15) and (16) with (17) also representing the branch voltage drop equation and the branch capacity equation after second-order cone relaxation is given in Equation (18). Equations (19–25) represent the PDN and PV operating constraints. The node voltage value constraint is modelled in Equation (19), whereas the branch current constraints are specified in Equation (20), and branch capacity limits are defined in Equation (21). Equations (22) and (23) represent the upper and lower limitations. Additionally, Equation (24) defines the reactive power limitation for PV, and Equation (25) sets the PV capacity constraint.

2.3 | TN Model Considering Flexible Swapping

Although spillback effects are critical in micro-traffic simulation, the static Wardrop equilibrium is adopted here to approximate the hourly average flow distribution for energy planning purposes, balancing modelling fidelity with the computational requirements of the coupled optimisation.

To facilitate the description of EV charging behaviours, the conventional traffic links are extended into four types of road links: conventional link T_A^{rg}, charging link T_A^{ch} (representing the

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EV's charging behaviour), battery swapping link T_A^{sp} (representing the EV's battery swapping behaviour) and bypass link T_A^{by} (representing vehicles passing charging stations without charging) [28]. These links are further expanded into regular arcs, charging arcs, swapping arcs and virtual arcs as shown in Figure 3 [13]. The travel time and cost of the bypass road links are assumed to be zero [29].

The TN model is constructed based on the Wardrop equilibrium model [30], and the road resistance model uses the Bureau of Public Roads function.

$$t_a^{rg}(x_a) = t_a^C\left[1 + 0.15\left(\frac{x_a^{rg}}{c_a^{rg}}\right)^4\right] : \forall a \in T_A^{rg} \tag{26}$$

The travel cost for a user is the sum of the travel costs of the road links they travel through, primarily including time costs and road congestion tolls. The total travel cost for GVs and EVs choosing charging or battery swapping modes denoted as $c_{k,rs}^g$, $c_{k,rs}^e$, $c_{k,rs}^s$ are defined as follows.

$$c_{k,rs}^g = \sum_{a\in T_A^{rg}} (\omega t_a^{rg} + T_a^C)\delta_{ak,rs}^g : k \in K_{rs}^g \tag{27}$$

$$\begin{aligned} c_{k,rs}^e = &\sum_{a\in T_A^{rg}} (\omega t_a^{rg} + T_a^C)\delta_{ak,rs}^e \\ &+ \sum_{a\in T_A^{ch}} \left[\omega t_a^{ch} + c_{e,t}\Delta SOC + M^{ch}\right]\delta_{ak,rs}^e : k \in K_{rs}^{ch} \end{aligned} \tag{28}$$

$$\begin{aligned} c_{k,rs}^s = &\sum_{a\in T_A^{rg}} (\omega t_a^{rg} + T_a^C)\delta_{ak,rs}^e \\ &+ \sum_{a\in T_A^{ch}} \left[\omega t_a^{sp} + c_{e,t}\Delta SOC + M^{sp}\right]\delta_{ak,rs}^e : k \in K_{rs}^{sp} \end{aligned} \tag{29}$$

After linearisation of Equations (27–29), the TN model considering flexible swapping is as follows:

$$\min C_{TN} = \sum_{rs}(\pi_{GV}u_{rs}^g + \pi_{ch}u_{rs}^e + \pi_{sp}u_{rs}^s)q_{rs} \tag{30}$$

$$\begin{aligned} \text{s.t.} x_a^{rg} = &\sum_{rs}\sum_{k\in K_{rs}^g} f_{k,rs}^g\delta_{ak,rs}^g + \sum_{rs}\sum_{k\in K_{rs}^e} f_{k,rs}^e\delta_{ak,rs}^e \\ &+ \sum_{rs}\sum_{k} f_{k,rs}^s\delta_{ak,rs}^s : a \in T_A^{rg} \end{aligned} \tag{31}$$

$$x_a^{ch} = \sum_{rs}\sum_{k} f_{k,rs}^e\delta_{ak,rs}^e : a \in T_A^{ch} \tag{32}$$

$$x_a^{sp} = \sum_{rs}\sum_{k} f_{k,rs}^s\delta_{ak,rs}^s : a \in T_A^{sp} \tag{33}$$

$$N_{GV} = \pi_{gv}q_{rs} = \sum_{k\in K_{rs}^g} f_{k,rs}^g \tag{34}$$

$$N_{ch} = \pi_{ch}q_{rs} = \sum_{k\in K_{rs}^e} f_{k,rs}^e \tag{35}$$

$$N_{sp} = \pi_{sp}q_{rs} = \sum_{k\in K_{rs}^s} f_{k,rs}^s \tag{36}$$

$$0 \le f_{k,rs}^g \perp c_{k,rs}^g - u_{rs}^g \ge 0 \tag{37}$$

$$0 \le f_{k,rs}^e \perp c_{k,rs}^e - u_{rs}^e \ge 0 \tag{38}$$

$$0 \le f_{k,rs}^s \perp c_{k,rs}^s - u_{rs}^s \ge 0 \tag{39}$$

Here, The notation $0 \le a \perp b \ge 0$ stands for complementarity and slackness conditions $a \ge 0$, $b \ge 0$ and $ab = 0$.

In detail, objective function includes three terms, which are the cost of GV users, EV users choosing charging and swapping, respectively. Equation (26) represents the calculation for travel time on regular links, whereas Equations (27–29) calculate the travel costs for GVs, charging EVs and swapping EVs, respectively. Equations (31–33) define the traffic flow conservation for conventional, charging and swapping links. Equations (34–36) specify the flow limits for GVs, charging EVs and swapping EVs choosing their respective paths. Equations (37–39) represent the complementarity constraints for GVs, charging EVs and swapping EVs within the network.

2.4 | The Coordination Optimisation Model

The power-transportation network is coupled through EV loads, and the coupling constraints are shown in Equation (40).

$$P_{load,j,t} = P_{load,j,t}^N + N_{a,ch}P_{load,j,t}^{ch} : a \in T_A^{ch} \tag{40}$$

Equation (40) represents the aggregate power demand within time interval t. We assume charging loads are uniformly distributed within the scheduling hour consistent with the steady-state traffic assumption.

The power-transportation coordination optimisation model is as follows [30] with the addition of constraint (40):

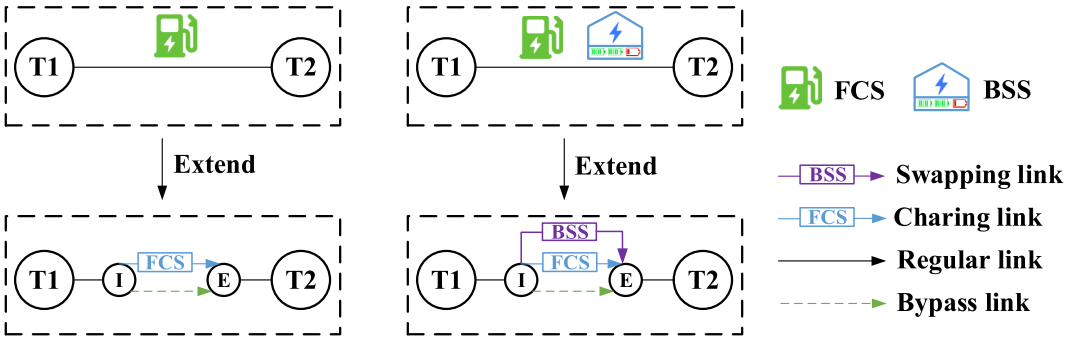


FIGURE 3 | Traffic link diagram model with CSS.

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$$\min\ C_{\mathrm{PDN}} + C_{\mathrm{TN}}$$

$$s.t \begin{cases} \text{Charging and swapping decision (1) - (13)} \\ \text{PDN constraints (15) - (25)} \\ \text{TN constraints (31) - (39)} \\ \text{Coupling constraints (40)} \end{cases} \tag{41}$$

3 | Solution Methodology

We note that the PDN model (14–25) and the TN model (30–39) are interrelated owing to the coupling constraints (40). However, the two models generally interact independently with limited information interchange. To simulate the interaction, on the one hand, we use an iterative algorithm to obtain equilibrium by sharing information on charging loads and prices.

On the other hand, to address the uncertainty of PV output, we propose a two-stage DRO framework. Unlike stochastic programming, which relies heavily on accurate probability distributions and robust optimisation, which tends to be overly conservative, the proposed DRO approach effectively balances these extremes. It utilises data-driven ambiguity sets to hedge against distribution errors while retaining computational tractability. Building upon the deterministic model, the framework first reduces M historical PV scenarios to K representative profiles with initial probabilities. Subsequently, an ambiguity set is constructed using 1-norm and ∞-norm constraints enabling robust optimisation against the worst-case probability distribution within the defined uncertainty bounds.

The first phase establishes day-ahead baseline operations by optimising deterministic model under forecasted PV scenarios. The second phase implements dynamic intra-day adjustments based on realised PV outputs, utilising a constructed ambiguity set with $1-$ norm and $\infty-$ norm constraints to guide real-time charging/swapping decisions through EV flow redistribution, thereby mitigating peak load [31]. The DRO model adopts a 'min–max–min' structure that systematically combines strategic day-ahead planning with adaptive intra-day responses.

The distributionally robust optimisation model of power-transportation systems is as follows:

$$\min_{x\in X}\left(Ax + \max_{\{p_s\}\in\psi}\sum_{s=1}^{N_s} p_s \min_{y_s\in Y_s} By_s\right) \tag{42}$$

$$Ax = \sum_{t=1}^{T} c_{\mathrm{sub},t}P_{\mathrm{sub},t} + \sum_{rs}\left(\pi_{\mathrm{ch}}u^{\mathrm{e}}_{rs} + \pi_{\mathrm{sp}}u^{\mathrm{s}}_{rs}\right)q_{rs} \tag{43}$$

$$By_s = \sum_{t=1}^{T}\left(\chi^{+}_{\mathrm{LE},t}P^{+}_{\mathrm{LE},t} + \chi^{-}_{\mathrm{LE},t}P^{-}_{\mathrm{LE},t}\right) \tag{44}$$

$$P_{\mathrm{LE},t} = P_{\mathrm{LE0},t} + P^{+}_{\mathrm{LE},t} - P^{-}_{\mathrm{LE},t} \tag{45}$$

$$g_{\mathrm{LE},t} + h_{\mathrm{LE},t} \le 1 \tag{46}$$

$$0 \le P^{+}_{\mathrm{LE},t} \le g_{\mathrm{LE},t}P^{+}_{\mathrm{LE,max}} \tag{47}$$

$$0 \le P^{-}_{\mathrm{LE},t} \le h_{\mathrm{LE},t}P^{-}_{\mathrm{LE,max}} \tag{48}$$

Equation (42) formulates the objective function, integrating multistage operational costs under PV uncertainty. The first component, expressed in Equation (43), captures the day-ahead economic scheduling costs dominated by electricity procurement expenditures through decision variable $P_{\mathrm{sub},t}$, u^{e}_{rs} and u^{s}_{rs}. Equation (44) quantifies the intra-day adjustment costs governed by variable $P^{+}_{\mathrm{LE},t}$ and $P^{-}_{\mathrm{LE},t}$, reflecting power load corrections required to reconcile deviations between forecasted and actual PV generation during real-time operations.

The operational constraints comprise two key components: Equation (45) governs controllable load adjustments and Equation (46) establishes a complementarity condition prohibiting concurrent load increase/decrease operations.

Considering the impact of PV output uncertainty, a confidence set can be constructed as follows:

$$\psi = \left\{ p \in R^{+}_{N} \middle| \begin{array}{l} \sum_{s=1}^{N_s} p_s = 1 \\ 0 \le p_s \le 1, \quad \forall\, s = 1, 2, \ldots, N_s \\ \sum_{s=1}^{N_s} |p_s - p_{s,0}| \le \theta_1 \\ \max\limits_{s=1,2,\ldots,N_s} |p_s - p_{s,0}| \le \theta_\infty \end{array} \right\} \tag{49}$$

$$\Pr\left\{\sum_{s=1}^{N_s}\left|p_s - p_{s,0}\right| \le \theta_1\right\} \ge 1 - 2N_s e^{-2K\theta_1/N_s} \tag{50}$$

$$\Pr\left\{\max_{1\le s\le N_s}\left|p_s - p_{s,0}\right| \le \theta_\infty\right\} \ge 1 - 2N_s e^{-2K\theta_\infty} \tag{51}$$

$$\theta_1 = \frac{N_s}{2K}\ln\frac{2N_s}{1-\alpha_1} \tag{52}$$

$$\theta_\infty = \frac{1}{2K}\ln\frac{2N_s}{1-\alpha_\infty} \tag{53}$$

The absolute value inequality constraints in Equation (49) exhibit inherent nonconvexity, which can be effectively addressed through auxiliary variable substitution. This standard linearisation technique reformulates the original constraints into equivalent linear expressions as follows.

$$\sum_{s=1}^{N_s} d \le \theta_1 \tag{54}$$

$$d \ge p_s - p_{s,0} \tag{55}$$

$$d \ge p_{s,0} - p_s \tag{56}$$

Figure 4 shows the summary of the model flow based on CCG algorithm. The CCG algorithm decomposes the primitive problem into a master problem (MP) and subproblem (SP), enabling iterative optimisation through their alternating solutions, which can be formulated as follows.

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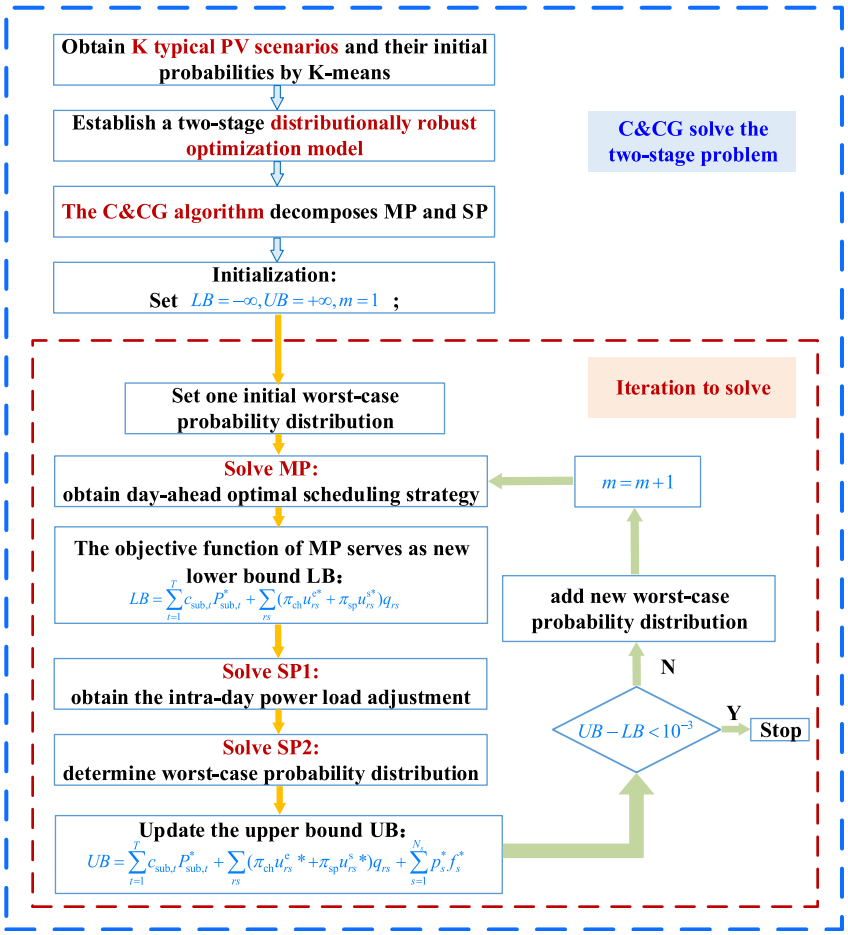


FIGURE 4 | A summary figure of the model flow based on CCG algorithm.

$$\mathrm{MP}: \min_{x\in X, y_s\in Y_s, \varphi} Ax+\varphi \tag{57}$$

$$\mathrm{SP}: f_{\mathrm{SP}}=\max_{p_s\in\psi}\sum_{s=1}^{N_s} p_s \min_{y_s\in Y_s} By_s: \forall s=1,2,\ldots,N_s \tag{58}$$

$$\mathrm{SP1}: \min_{y_s\in Y_s} f_s=By_s: \forall s=1,2,\ldots,N_s \tag{59}$$

$$\mathrm{SP2}: \max_{p_s\in\psi}\sum_{s=1}^{N_s} p_s f^{*}_{s}, : \forall s=1,2,\ldots,N_s \tag{60}$$

4 | Example

Figure 5 shows the topology, which includes a three-bus PDN and a five-node TN. The rated voltage of the PDN system is 12.66 kV with node 1 connected to the main grid. The base power is 1 MVA, and the base voltage is 10 kV. The base power is 1 MVA, and the base voltage is 10 kV. The voltage limits are 1.1 and 0.9 p.u. respectively. Considering actual behaviours, the initial BRP is set to 0.8.

The baseline traffic flow of TN is 500 vehicles. The unit time cost is 10 $/hour. Other data such as capacity of every link and free travel time are available through an online website.[1]

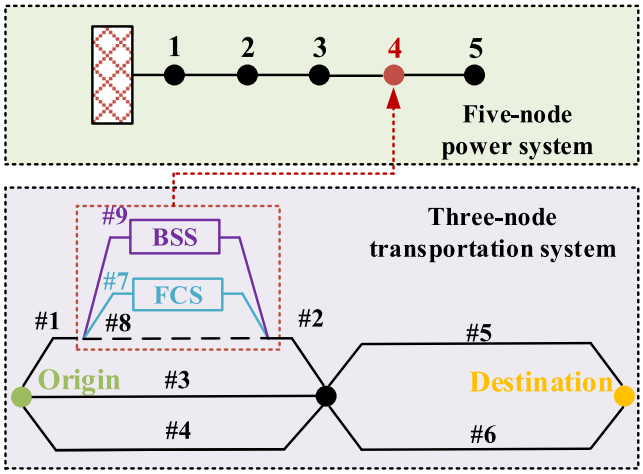


FIGURE 5 | Coupled five-bus power system and three-node transportation system.

The optimisation is programmed in GAMS 24.7 and solved by BARON and GUROBI on a personal computer with R9-7945HX processor with 16 GB of memory.

The NLP solution reveals the relationship between traffic flow distribution and operational costs in Figure 6. Traffic volume and charging costs jointly drive operational cost variations: as

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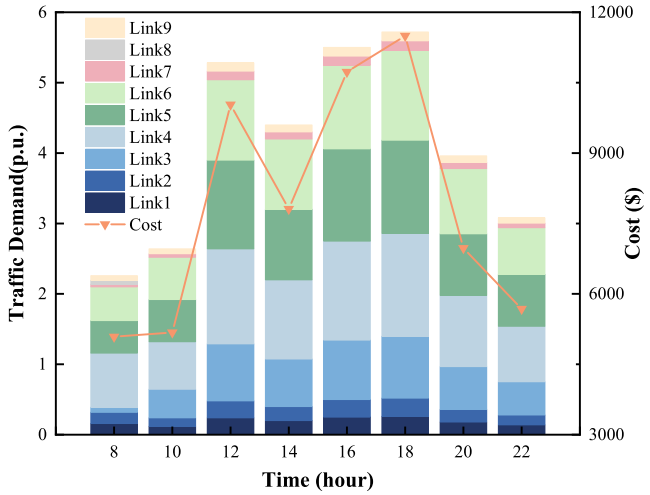


FIGURE 6 | Traffic flow distribution and cost of 3–5 node topology.

both factors increase, total costs rise accordingly. This trend closely follows the traffic demand pattern shown online with operational costs reaching their peak of $11,496.76 at the 18th hour when traffic volume peaks. Meanwhile, conventional road segments (Links 3–6) maintain relatively high traffic flow since most vehicles do not require energy replenishment during these routes.

To further analyse users' charging and swapping preferences, we extract charging flow (link 7) and battery swapping flow (link 9) data and calculate the battery swapping probability as shown in Figure 7. Under the assumption that EVs can swap batteries, the charging and swapping preferences of EV users are mainly influenced by energy replenishment costs and replenishment time. During low-traffic periods with time-critical travel needs, the time-saving advantage of battery swapping outweighs its higher service fees driving increased adoption. Conversely, mid-afternoon periods with flexible schedules exhibit reduced swapping demand as users prioritise lower-cost charging options. This decision pattern empirically validates Wardrop's first principle, confirming that EV users systematically select energy replenishment strategies to optimise personal cost-time tradeoffs.

The simulation outcomes conclusively demonstrate the operational viability of power-transportation nexus model while establishing the behavioural feasibility of context-aware charging/swapping station selection mechanisms. These dual validations provide critical methodological foundations for advancing next-generation transportation electrification architectures with scalable vehicle-grid integration capabilities.

5 | Case Study

This section illustrates the performance of the proposed model using a case study involving IEEE 33-bus system and 13-node TN that is shown in Figure 8. The three FCSs are located at nodes #22, #25 and #28 whereas the three BSSs are located at nodes #29, #30 and #31. The other data are available through an online supplement. Other data are also available through the same website in example.

5.1 | Flow Distribution of Power-Transportation Coordination Model

The coordinated optimisation solution for nodes 13–33 generates traffic flow distributions shown in Figure 9, exhibiting spatiotemporal patterns aligned with empirical urban mobility characteristics. Peak congestion manifests between 12:00 and 16:00, and critical corridors (Links 4, 14, 15, 19, 23 and 24) demonstrate significantly higher throughput. These findings deliver actionable insights for infrastructure operators and transportation planners to strategically formulate time-dependent traffic routing policies and dynamic charging load allocation mechanisms, thereby alleviating network congestion while ensuring energy demand–supply equilibrium.

Figure 10 shows operational flow analysis for the three CSSs. During low-traffic periods, FCS 2 attracts significantly higher user preference than Stations 1 and 3 due to its lower electricity pricing. In high-traffic scenarios, however, queue delays and congestion-induced travel costs offset the price advantages of individual stations, leading to balanced EV distribution across all three stations with minimal inter-station usage disparity.

The simulation outcomes conclusively demonstrate the power-transportation coordination model maintains operational efficacy within the 13–33 node network topology. Besides, it still maintains a relatively high computational speed and average calculating time is 59.6s, whereas the 3–5 node network is 14s. These results empirically validate the structural scalability and multinetwork compatibility of the proposed coordination framework, delivering implementable insights for real-world power-transportation system integration.

5.2 | Comparison of Power-Transportation Coordination Model Considering CSS or Only FCS

A comparative analysis of power-transportation coordination optimisation is conducted for the 13–33 node network under two configurations: integrated charging–swapping stations versus fast-charging-only stations as shown in Figure 11. Figure 11a

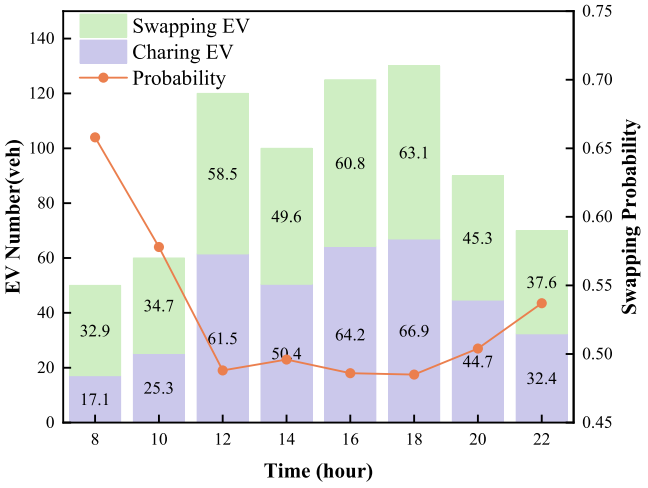


FIGURE 7 | Charging/swapping EV numbers and swapping probability in 3–5 node topology.

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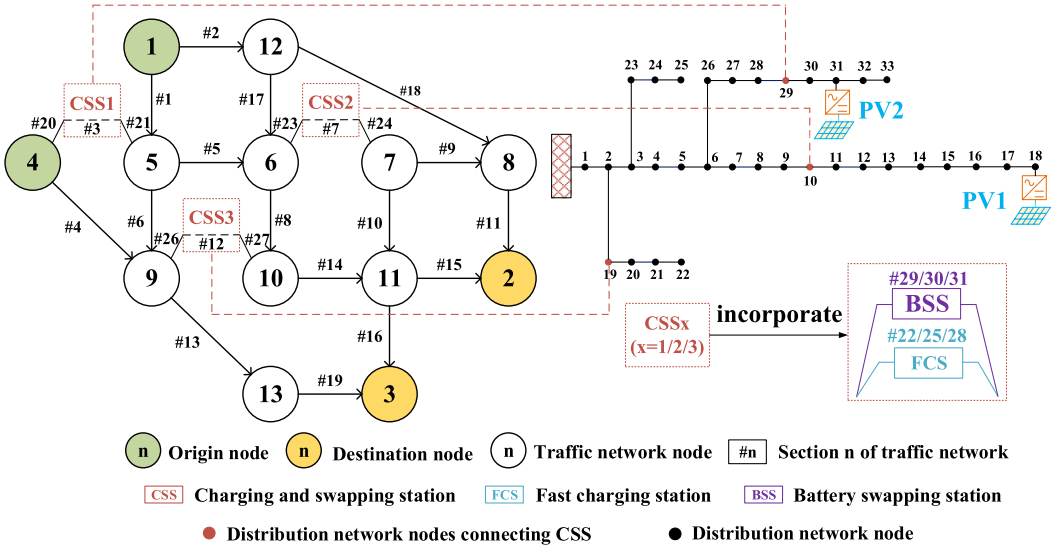


FIGURE 8 | Coupled 13-node transportation system and modified IEEE 33-bus system with high renewable penetration.

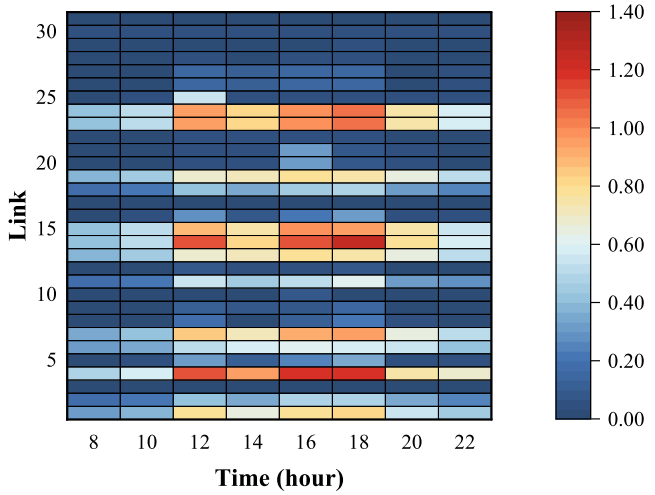


FIGURE 9 | Traffic network flow distribution of 13–33 node topology.

illustrates the voltage distribution profiles under the integrated charging–swapping configuration, whereas Figure 11b depicts the fast-charging-only scenario.

The fast-charging-only configuration restricts users to a single charging modality, leading to exacerbated voltage security challenges during peak demand periods. This operational constraint results in recurrent voltage violations particularly more frequent breaches of the lower voltage limit as shown in Figure 11b. Conversely, the integrated charging–swapping system demonstrates superior voltage regulation, maintaining profiles predominantly within the relative security envelope. This configuration sustains compliant voltage levels across 80% of network nodes during peak intervals compared to 64.8% in fast-charging-only systems. The integrated approach reduces voltage limit violation risks by 15% during critical demand windows.

Figure 12 compares the service duration, encompassing both queueing and charging times, between integrated charging-swapping stations and fast-charging-only stations across nodes 13–33. The fast-charging-only configuration requires an average peak-period service duration of 182.05 min. In contrast, integrated stations achieve a significantly reduced average service time of 78.87 min by offering battery swapping as an alternative energy replenishment option. This dual mode approach reduces the total energy replenishment time by 56.7%, improves the efficiency of the system throughput by a factor of 2.3 and greatly improves user satisfaction compared to conventional fast charging systems.

The simulation outcomes conclusively demonstrate the marked operational superiority of the power-transportation coordination model over conventional frameworks particularly in enhancing grid voltage stability and minimising user service durations. These findings deliver actionable insights for addressing critical challenges in transportation electrification: mitigating EV range anxiety through adaptive energy replenishment strategies and alleviating grid peak-load pressures during extreme weather conditions. The empirical validation of the role of flexible swapping emphasises its indispensability in advancing cross-domain power-transportation system optimisation.

5.3 | DRO Results of Power-Transportation Coordination Model

The 24-h operational cost analysis of three CSSs validate the feasibility of DRO model, utilising 117 PV output scenarios processed via fuzzy *K*-means clustering into 10 distinct operational modes. For deterministic model, averaging EV mobility costs across representative time slots determine the optimal day-ahead scheduling benchmark. The concrete data are available through an online supplement.

Table 1 comparatively quantifies the electricity procurement costs and EV mobility expenditures under varying confidence intervals. The results reveal a positive correlation between confidence interval magnitude and systemic risk-aversion measures. Expanded confidence levels intensify PV output uncertainty, necessitating

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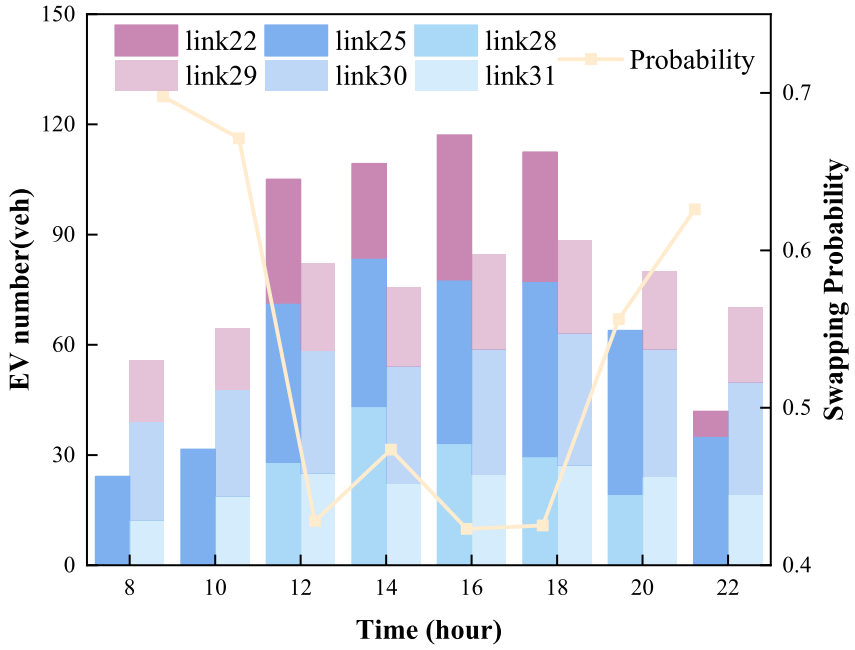


FIGURE 10 | Traffic flow distribution and cost of 13–33 node topology.

increased reserve electricity procurement that elevates total operational costs. This computational alignment with the cost escalation principle formulated in Equations (52) and (53) confirms the model's mathematical rigour, while demonstrating practical tradeoffs between operational robustness and economic efficiency in power-transportation systems.

Furthermore, with θ_1 fixed at 0.5, we compare the outcomes under the composite norm constraint for $\theta_\infty = 0.5/0.9/0.95$ to those under the $1-$ norm constraint with the same θ_1. The analysis, presented in Table 2, reveals that the composite norm constraint, by incorporating both the $1-$ norm and $\infty-$ norm, exhibits lower conservativeness than the 1-norm constraint alone. Consequently, the total cost under the composite norm constraint is reduced compared to that under the 1-norm constraint.

The results quantify the impact of variations in PV generation on power-transportation system. Accounting for PV generation uncertainty in real-time operations enables more precise intraday cost estimation, offering valuable insights for CSS operators and grid schedulers. Through this hierarchical structure, the model achieves three key synergies. Firstly, it mitigates probabilistic risks associated with PV variability using DRO techniques. Secondly, it achieves spatiotemporal load balancing through optimised EV routing and charging coordination. Finally, it reduces infrastructure stress by strategically decoupling peak charging demands from periods of low PV generation.

5.4 | Sensitivity Analysis

In the previous work, we set BRP to a fixed value, and here, we change its value to evaluate its influence on battery swapping probability and operational costs. Figure 13 presents the results. Figure 13a illustrates the relationship between BRP and battery

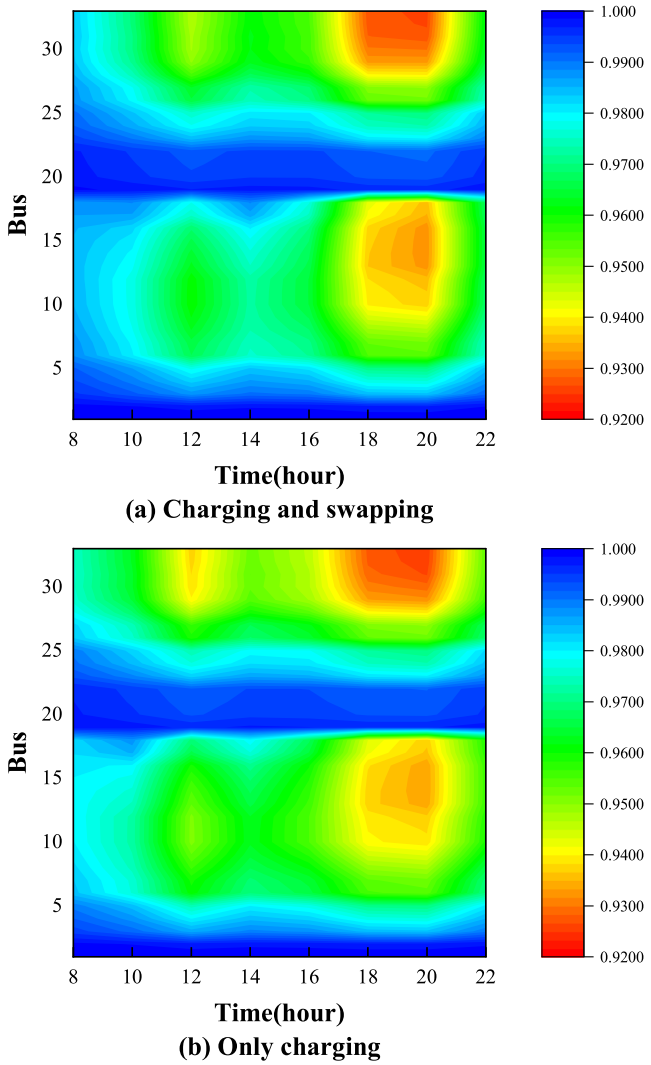


FIGURE 11 | Comparison of voltage distribution between CSS and only FCS.

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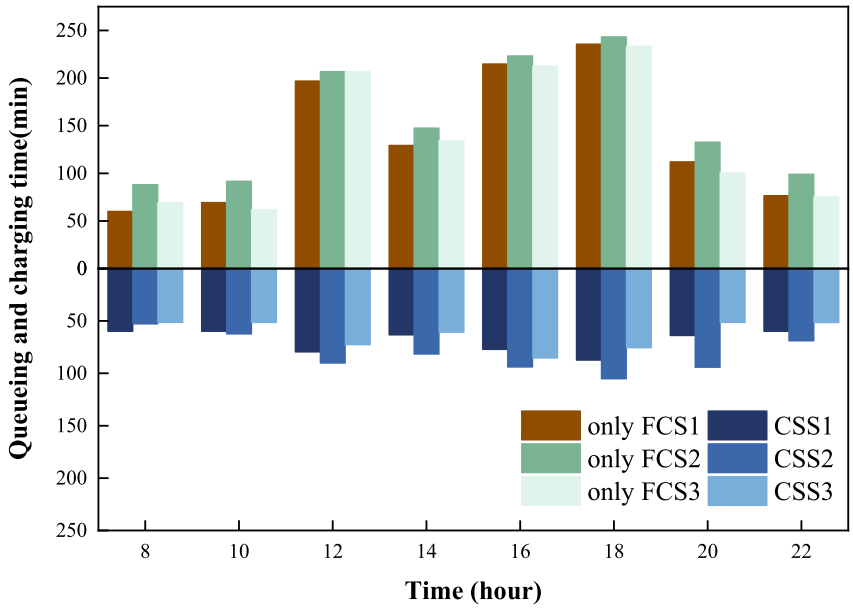


FIGURE 12 | Comparison of queueing and charging time between CSS and only FCS.

TABLE 1 | Results comparison under different confidence sets.

	Operating cost ($)		
θ_1	$\theta_\infty = 0.5$	$\theta_\infty = 0.9$	$\theta_\infty = 0.95$
0.2	5792.834	5787.622	5791.365
0.5	5792.834	5787.622	5791.365
0.95	5792.834	5787.622	5791.365

TABLE 2 | Comparison of comprehensive norm and 1 − norm results.

	Operating cost ($)	
θ_∞	Composite norm	1 − norm
0.2	5792.834	5813.1
0.5	5787.622	5813.1
0.95	5791.365	5813.1

swapping probability. As the parameter increases, the swapping probability stabilises. When the parameter approaches 1, the probability initially decreases and then increases. Conversely, as the parameter nears 0, the rational decision-making capacity of EV users diminishes, resulting in greater fluctuations in their preferences for charging versus swapping. Further analysis reveals that under low traffic flow conditions, EVs exhibit a higher probability of selecting battery swapping, whereas during mid-afternoon hours, the probabilities of choosing charging or swapping are approximately equal.

Figure 13b depicts the relationship between the BRP and operational costs. The cost trend generally correlates with traffic flow but displays fluctuations particularly during midday and evening periods. Variations in traffic flow and BRP amplify cost fluctuations during these times, whereas costs remain relatively stable during morning and midnight hours.

As BRP approaches 1, users become highly sensitive to cost differences, leading to a rapid convergence towards the mathematically optimal choice. This 'phase transition' behaviour reflects the sharp reduction in perception error.

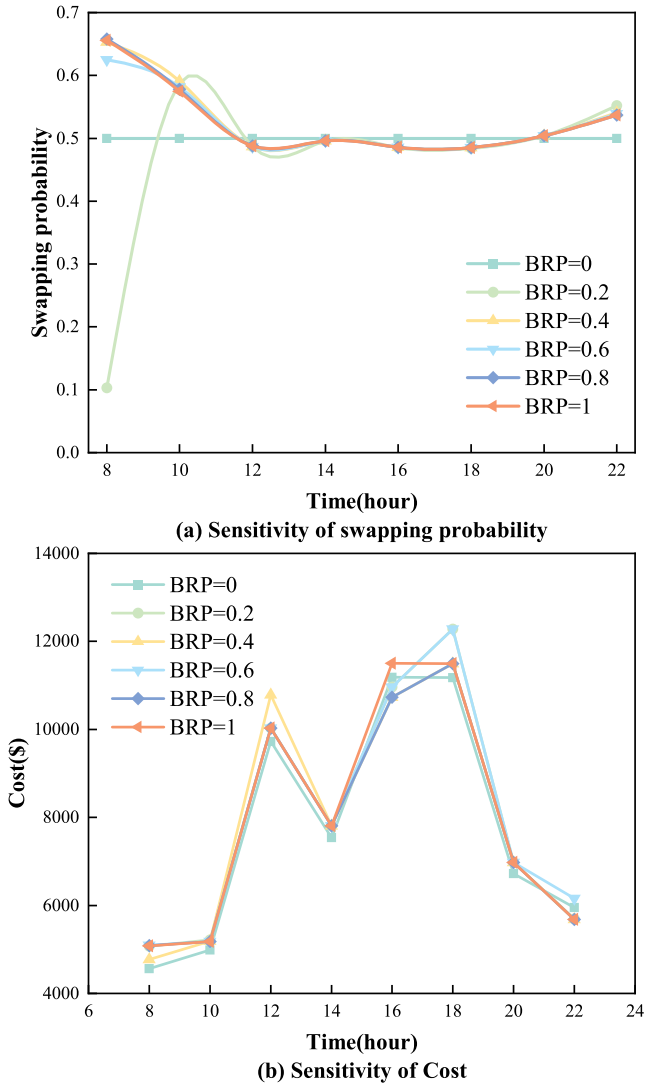
The sensitivity analysis results demonstrate how BRP affects the battery swapping probability and operational costs, indicating that BRP plays an important role in the charging and swapping choices of EV users as well as in operational costs. Therefore, when guiding the charging and battery swapping behaviour of EVs, it is crucial to fully consider the impact of BRP in order to make more comprehensive subsequent scheduling plans.

Besides, sensitivity analysis on the swapping penetration rate (α) reveals a nonlinear threshold effect. Although low penetration ($\alpha \leq 0.6$) fails to trigger coordination benefits, a high penetration level (e.g., $\alpha = 0.9$) yields a significant cost reduction to \$4760.8, outperforming the full-penetration baseline ($\alpha = 1.0$) by mitigating queueing congestion at swapping stations through the natural diversion of nonswappable EVs to charging facilities.

5.5 | Practical Implications

The above results show that the introduction of flexible swapping and user stochastic equilibrium model greatly alleviates the user's energy replenishment time, reduces the peak load bearing pressure of the power grid and effectively reduces the risk of voltage over-limit. At the same time, the established power-transportation coordination model also shows satisfactory calculation speed and convergence accuracy, and the average solution time of the model is about 69.5s. In the future, more large-scale power-transportation coordination topologies will be further considered to provide guidance for CSS operators. In addition, the setting of BRP currently relies only on experience. In the future, it will be further considered to extract BRP from

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(a) Sensitivity of swapping probability

(b) Sensitivity of Cost

FIGURE 13 | Sensitivity analysis of BPR, battery replacement probability and operating cost.

actual user behaviour under different spatial and temporal distributions to collaboratively optimise the distribution of charging and swapping behaviour.

6 | Conclusion

This paper proposes a power-transportation coordination optimisation model integrating battery swapping services to address range anxiety issue. The model also presents a novel solution for mitigating peak electricity demand during summer periods through battery swapping mechanisms. The proposed framework employs the SUE model and BRP to determine the probability distribution of EV users' preferences between charging and battery swapping options. Furthermore, DRO is implemented to achieve power-transportation coordination optimisation considering PV output fluctuations.

Our work provides forward-looking insights for future power-transportation coordination systems incorporating battery swapping technologies, which will effectively alleviate the user range anxiety and the risk of power grid over-limit. Our work also suggests operational departments of both PDN and TN have substantial incentives to share operational information, thereby improving system safety and operational efficiency.

Admittedly, information barriers and data privacy constraints still exist between the two networks in practice. However, in the future, these challenges can be addressed through third-party aggregators or privacy-preserving communication protocols where only marginal price/flow signals are exchanged without revealing sensitive user data.

Nomenclature

Abbreviations

BPR	Bounded rationality parameter
BSS	Battery swapping station
CCG	Column and constraint generation
CEL	controllable electrical load
CSS	Charging–swapping station
DRO	Distributionally robust optimisation
EV	Electric vehicle
FCS	Fast charging station
GV	Gasoline vehicle
OD	Origin-Destination
PDN	Power distribution network
PV	Photovoltaic
SUE	Stochastic user equilibrium
TN	Transportation network

Indices and sets

a	Index of traffic network links
ij	Index of branch
i,j	Indices of PDN buses
t	Index of time intervals
K_{rs}^{g}	Set of GVs paths
K_{rs}^{ch}	Set of charging EVs paths
K_{rs}^{sp}	Set of swapping EVs paths
$N(j)$	Set of terminal nodes of branches
T	Set of time intervals
$T(i)$	Set of head-end nodes of branches
T_A^{rg}	Set of regular road links
T_A^{ch}	Set of charging road links
T_A^{sp}	Set of swapping road links
T_A^{by}	Set of bypass road links
Y_s	Set of second-stage decision variables
ψ	Set of confidence for a probability distribution

Parameters

$\alpha_{1/\infty}$	1/∞-norm confidence

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A/B	Vector or matrix form of the coefficients
$c_a^{\mathrm{ch/sp}}$	EV capacity of FCS/BSS (veh)
c_a^{rg}	Traffic flow capacity on regular link$a$ (veh)
$c_{e,t}$	Charging price per unit during period t (\$/kWh)
$c_{\mathrm{sub},t}$	Electricity purchase price during period t (\$/kWh)
E	Energy demand of a single EV (veh)
$I_{ij,\min}$	Minimum current limitation on branch ij (p.u.)
$I_{ij,\max}$	Minimum current limitation on branch ij (p.u.)
K	Number of historical data points
$M_{\mathrm{ch/sp}}$	Charging/swapping service fee (\$/kWh)
$N_{\mathrm{ch/sp}}$	Number of charging/swapping EVs (veh)
N_{GV}	Total number of GVs (veh)
N_{EV}	Total number of EVs (veh)
N_s	Total number of PV scenarios
p	Charging power of a single EV (p.u.)
p_s	Probability of the occurrence of the $s-th$ scenario
q_{rs}	Overall travel demand between the OD pair (veh)
$P_{\mathrm{sub}}^{\min}$	Minimum active power (p.u.)
$P_{\mathrm{sub}}^{\max}$	Maximum active power (p.u.)
$P_{j,t}^{\mathrm{load}}$	Load power at j node at time t (p.u.)
$P_{N,j,t}^{\mathrm{load}}$	Fundamental load power at j node at time t (p.u.)
$P_{\mathrm{LE0},t}$	Base electrical load at time t (p.u.)
$P_{j,t}^{\mathrm{PV}}$	PV active power at j node at time t (p.u.)
$Q_{j,t}^{\mathrm{PV}}$	PV reactive power at j node at time t (p.u.)
$Q_{\mathrm{sub}}^{\min}$	Minimum reactive power limitation (p.u.)
$Q_{\mathrm{sub}}^{\max}$	Minimum reactive power limitation (p.u.)
$Q_{\mathrm{PV}}^{\min}$	Minimum reactive power output of PV (p.u.)
$Q_{\mathrm{PV}}^{\max}$	Maximum reactive power output of PV (p.u.)
$S_{\mathrm{PV}}^{\max}$	Maximum capacity of PV (p.u.)
$S_{ij,\max}$	Maximum capacity of branch ij (p.u.)
$t_{a,\mathrm{ch/sp}}^{\max}$	Queueing time at full capacity of the FCS/BSS (min)
$t^{\mathrm{ch/sp}}$	Charging/swapping duration of EVs (min)
t_a^C	Free-flow travel time on regular link a (min)
$U_{i,\min}$	Minimum voltage limitation at node i (p.u.)
$U_{i,\max}$	Maximum voltage limitation at node i (p.u.)
$\pi_{\mathrm{ch/sp}}$	Penetration rate of charging/swapping EVs (%)
π_{GV}	Penetration rate of GVs (%)
π_{EV}	Penetration rate of EVs (%)
ω	Travel cost (\$/min)
λ	Bounded rationality parameter
$\chi_{\mathrm{LE},t}^{+/-}$	Upward/downward adjustment cost coefficient of CEL
$\theta_{1/\infty}$	Probability allowable deviation limit under $1/\infty$-norm constraint

Variables

$\alpha_{\mathrm{ch},n}$	Probability of travelling to FCS n
$\alpha_{\mathrm{sp},n}$	Probability of travelling to BSS n
$c_{k,rs}^{\mathrm{g/e/s}}$	Total travel cost of GVs or charging/swapping EVs (\$)
$\tilde{C}_{\mathrm{ch/sp}}$	Random charging/swapping cost (\$)
$C^{\mathrm{ch/sp}}$	Total charging/swapping cost of a single EV (\$)
C_{PDN}	Total operating cost of the PDN (\$)
C_{TN}	Total operating cost of the TN (\$)
C_{sub}	Electricity purchasing cost (\$)
$f_{k,rs}^{\mathrm{g/r/e}}$	Traffic flow of GVs or charging/swapping EVs choosing path k (p.u.)
$g_{\mathrm{LE},t}$	Upward adjustment state variable of CEL
$h_{\mathrm{LE},t}$	Downward adjustment state variable of CEL
$I_{ij,t}$	Current value on branch ij (p.u.)
$l_{ij,t}$	Square of the current value on branch ij (p.u.)
$N_{a,\mathrm{ch}}$	Number of EVs in the FCS (veh)
$N_{a,\mathrm{sp}}$	Number of EVs in the BSS (veh)
$P_{\mathrm{sub},t}$	Power purchased at node j at time t (p.u.)
$P_{ij,t}$	Active power on branch ij at time t (p.u.)
$P_{j,t}^{\mathrm{load}}$	Total load power at j node at time t (p.u.)
$P_{\mathrm{ch},j,t}^{\mathrm{load}}$	Charging load power at node j at time t (p.u.)
$P_{\mathrm{LE},t}^{+/-}$	Upward/downward adjustment power of CEL (p.u.)
$Q_{\mathrm{sub},t}$	Reactive power at time t (p.u.)
$Q_{ij,t}$	Reactive power on branch ij at time t (p.u.)
$Q_{j,t}^{\mathrm{PV}}$	Reactive power output of PV at node$j$ at time t (p.u.)
t_a^{rg}	Travel time on regular link a (min)
$t_a^{\mathrm{ch/sp}}$	Total charging/swapping time at FCS/BSS (min)
T_a^C	Congestion toll on link a (\$/min)
$u_{rs}^{\mathrm{g/e/s}}$	Minimum travel cost of GVs or charging/swapping EVs between OD pair $r-s$ (\$)
v_{rs}^k	Binary variable for each path
$V_{i,t}$	Square of voltage value at node i (p.u.)
x	First-stage decision variables
x_a^{rg}	Traffic flow on regular link $a \in T_A^{\mathrm{rg}}$ (veh)
x_a^{ch}	Traffic flow on charging link $a \in T_A^{\mathrm{ch}}$ (veh)
x_a^{sp}	Traffic flow on swapping link $a \in T_A^{\mathrm{sp}}$ (veh)
y_s	Second-stage variables associated with scenario s
$\tan\theta_j$	Voltage angle at node j (p.u.)
$\varepsilon_{\mathrm{ch},n}$	Stochastic term of charging cost
$\varepsilon_{\mathrm{sp},n}$	Stochastic term of swapping cost
$\delta_{ak,rs}^{\mathrm{g/e}}$	If the link a belongs to the path k for GVs/EVs (yes: 1, no: 0), indicating the coupling relationship between the regular link and the path
ϖ_a	Vector where at most two adjacent variables can be nonzero

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Conflicts of Interest

The authors declare no conflicts of interest.

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Data Availability Statement

Some data supporting this study are included in the article and others will be available in Figshare at http://doi.org/10.6084/m9.figshare.28691747.

Endnotes

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